

SOM2IMG: Self-Organizing Map Based Tabular-to-Image Transformation for Deep Learning

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Abstract—Tabular-to-image (T2I) methods convert tabular data into image representations, enabling the use of convolutional neural networks (CNNs), yet the relative impact of feature placement strategies on model performance remains insufficiently understood. We propose SOM2IMG, a T2I method that trains a Self-Organizing Map (SOM) on the transpose of the data matrix to learn a two-dimensional layout in which similar features are placed nearby; each sample is then represented as an image by mapping its feature values onto this fixed grid. We evaluate SOM2IMG and an expanded-grid variant against six existing T2I methods and two control baselines, fitting all transformations on training data and using stratified 10-fold cross-validation across twelve datasets spanning eight domains. Statistical analysis identifies a small set of strong feature preserving methods, namely SOM2IMG, correlation based mapping, and IGTD, that consistently outperform random baselines, with SOM2IMG achieving the best overall performance. In contrast, methods that discard features, including DeepInsight, REFINED, and SuperTML perform worse than even a random placement baseline. No single method dominates across datasets, and because the sample of twelve datasets is small, these differences do not reach statistical significance; the strongest methods are therefore best treated as a set worth trying. A controlled ablation study further shows that preserving all features in the image representation is the primary factor associated with strong performance, while spatial arrangement provides additional benefit mainly in higher-dimensional settings. These results demonstrate that SOM2IMG is a competitive and interpretable T2I approach and establish a practical design principle for T2I methods: retaining all features is more critical than the specific placement strategy.

Index Terms—Self-Organizing Maps, Tabular-to-Image Transformation, Convolutional Neural Networks, Tabular Data, Deep Learning, Critical-Difference Analysis

I. INTRODUCTION

CNNs do very well on data with real spatial structure, such as images and audio, because they reuse local patterns and stack them into a hierarchy [1], [2]. This works because of a built-in assumption: neighboring pixels are related, which lets the filters pick up edges, textures, and larger shapes [3], [4]. Tabular data breaks that assumption. The columns of a table are usually separate measurements in an arbitrary order, so adjacency carries no semantic meaning. A filter sliding over unrelated columns provides limited exploitable structure [5], [6], which is a big part of why gradient boosting [7], [8] and random forests [9] still beat CNNs on tabular data.

T2I methods aim to overcome this limitation by presenting the columns as an image, enabling the application of a CNN to tabular data. Techniques such as DeepInsight [10] and IGTD [11] employ distinct approaches, including dimensionality reduction or correlation clustering, to group related features close to each other. However, most of these methods either require the labels or lack a principled method to determine which features truly belong together.

SOM2IMG takes a different approach. Instead of treating each sample as a point, it trains a SOM on the transpose of the data, so each feature becomes a point described by its values across all samples. The SOM then groups similar features on a 2D grid without labels, and that grid becomes the template for the image. Each feature is placed at its Best Matching Unit (BMU) with its value left untouched, keeping the layout readable and the data preserved.

The evaluation protocol matters in T2I settings, since the transformation itself is learned from data. We fit each layout (and every normalization step) on the training fold only; otherwise the held-out folds would shape the layout and inflate the scores. Comparing methods with the standard many-dataset test [12], we find several are statistically tied, and the clearest distinction between effective and ineffective methods is simply whether they retain all the features.

Our contributions are: (1) We propose SOM2IMG, a tabular-to-image transformation that learns a topology-preserving feature layout using a Self-Organizing Map trained in feature space, along with an expanded-grid variant that reduces feature collisions; (2) we conduct a controlled evaluation across twelve datasets using stratified cross-validation and statistical testing (Friedman test with Holm-corrected post-hoc analysis), ensuring that all transformations are fit only on training data; (3) we show that SOM2IMG performs competitively with state-of-the-art T2I methods, forming a statistically indistinguishable top group and achieving the best performance on multiple datasets; and (4) we demonstrate that preserving all features is the primary factor associated with strong T2I performance, and that SOM2IMG provides a fully interpretable, value-preserving mapping in which each pixel corresponds to an original feature.

II. RELATED WORKS

A. Self-Organizing Maps

Self-Organizing Maps (SOMs) have been used for decades for unsupervised learning and visualization [13]. They preserve local neighborhoods, provide a readable 2D map, and naturally group similar items together, making them well-suited for T2I, where spatial coherence is crucial. Unlike dimensionality reduction, which can distort local neighborhoods, or optimization methods, which can get stuck in suboptimal configurations, a SOM offers a clean way to arrange data while preserving both local and global structure.

SOMs have been used in conjunction with CNNs before, but for different tasks. For instance, Agboka et al. [14] employed a SOM before a CNN for malaria-vector resistance, while Ramirez-Quintana & Chacon-Murga [15] utilized one for object detection. In both cases, the SOM clustered samples rather than arranging features into a grid for the CNN to read. To our knowledge, prior work has not applied SOMs to create feature-space layouts for T2I transformations.

B. Existing T2I Algorithms

Several T2I methods exist and have been benchmarked [16]. The ones we compare against are listed below.

- **DeepInsight** projects the features into 2D with t-SNE or kernel PCA and drops each feature at the nearest pixel [10].
- **IGTD** treats placement as an optimization that preserves the rank order of feature distances [11].
- **REFINED** uses multidimensional scaling plus hill-climbing to nudge correlated features together [17].
- **NCTD** expands the features into a $2N \times 2N$ image of pairwise interactions [18]; that quadratic size gets expensive fast as the feature count grows.
- **SuperTML** writes feature values as glyphs at positions chosen by supervised feature selection [19].
- **Correlation-Based mapping** orders features by hierarchical clustering of their correlations, then lays them along a space-filling path so related features end up as neighbors [20].

These algorithms cover dimensionality reduction, optimization, and correlation heuristics, but they face challenges in terms of interpretability, the need for labels, and computational cost, which motivated our SOM approach.

III. METHODS AND MATERIALS

A. Datasets

We use the twelve datasets in Table I. Seven are public sets [21] covering medical diagnosis, bean morphology, and spam; an eighth, the Heart and Soul study, is a private cohort of 515 coronary-heart-disease patients with about 100 clinical variables. To broaden domain and feature-range coverage we add four more: German Credit (finance), Ionosphere (radar), MADELON (a synthetic feature-selection set that is mostly noise), and a single-cell mouse hepatoblast genomics set reduced to its top principal components. Together they span

TABLE I
DATASETS USED IN THE EVALUATION. N = SAMPLE SIZE; d = FEATURE COUNT AFTER PREPROCESSING; C = NUMBER OF CLASSES.

Dataset	N	d	C	Domain
Breast Cancer Wisconsin	569	30	2	Medical
Heart Disease	303	20	2	Medical
Lung Cancer	32	101	3	Medical
Myocardial Infarction	1700	184	8	Medical
Heart and Soul	515	100	2	Medical
Obesity Levels	2111	23	7	Lifestyle
Dry Beans	13611	16	7	Agriculture
Spambase	4601	57	2	Text / spam
German Credit	1000	48	2	Finance
Ionosphere	351	34	2	Radar
MADLON	2600	500	2	Synthetic
Mouse hepatoblast	447	447	2	Genomics

$d = 16$ to $d = 500$ features, mix binary and multiclass targets, and cover eight domains.

B. SOM2IMG: Transformation

SOM2IMG employs a SOM trained on features, not samples, to lay features on a 2D grid. Features that exhibit similar behavior across the dataset tend to cluster together. A sample is transformed into an image by dropping its feature values at the grid spots corresponding to those features. Figure 1 illustrates this process. There are three distinct phases involved.

Phase 1: Represent the features. For data $X \in \mathbb{R}^{n \times d}$ with n samples and d features, we min-max scale each feature (using training-fold statistics only; see Sec. III-E) and transpose it to $X^T \in \mathbb{R}^{d \times n}$, so each feature is now a vector of its values over the training samples.

Phase 2: Train the SOM. We build a 2D grid of size $h \times w$,

$$h = \lceil \sqrt{cd} \rceil, \quad w = \left\lceil \frac{cd}{h} \right\rceil, \quad (1)$$

where $c \geq 1$ is a grid-expansion factor ($c = 1$ for the default SOM2IMG and $c = 3$ for the expanded SOM2IMG-T). Training sends each feature to its Best Matching Unit (BMU),

$$\text{BMU}(f_i) = \arg \min_j \|f_i - w_j\|_2, \quad (2)$$

where f_i is the i -th feature vector and w_j is the weight vector of neuron j . Similar features settle next to each other, giving a topology-preserving map. We use the MiniSom implementation on the grid from Eq. 1, with a Gaussian neighborhood (starting radius $\sigma=1.0$), a starting learning rate of 0.5, and 5,000 random training steps. We initialize SOM weights randomly and use a fixed random seed for reproducibility.

Phase 3: Make the images. Each feature is assigned to a BMU cell (as defined in Eq. 2). However, since the BMU map is many-to-one, features can collide on a single neuron. Instead of overwriting them, we iteratively assign the colliding feature to the nearest available grid cell, determined by the Euclidean distance. This ensures each feature is assigned exactly once. The remaining cells are set to zero. Finally, a sample's image

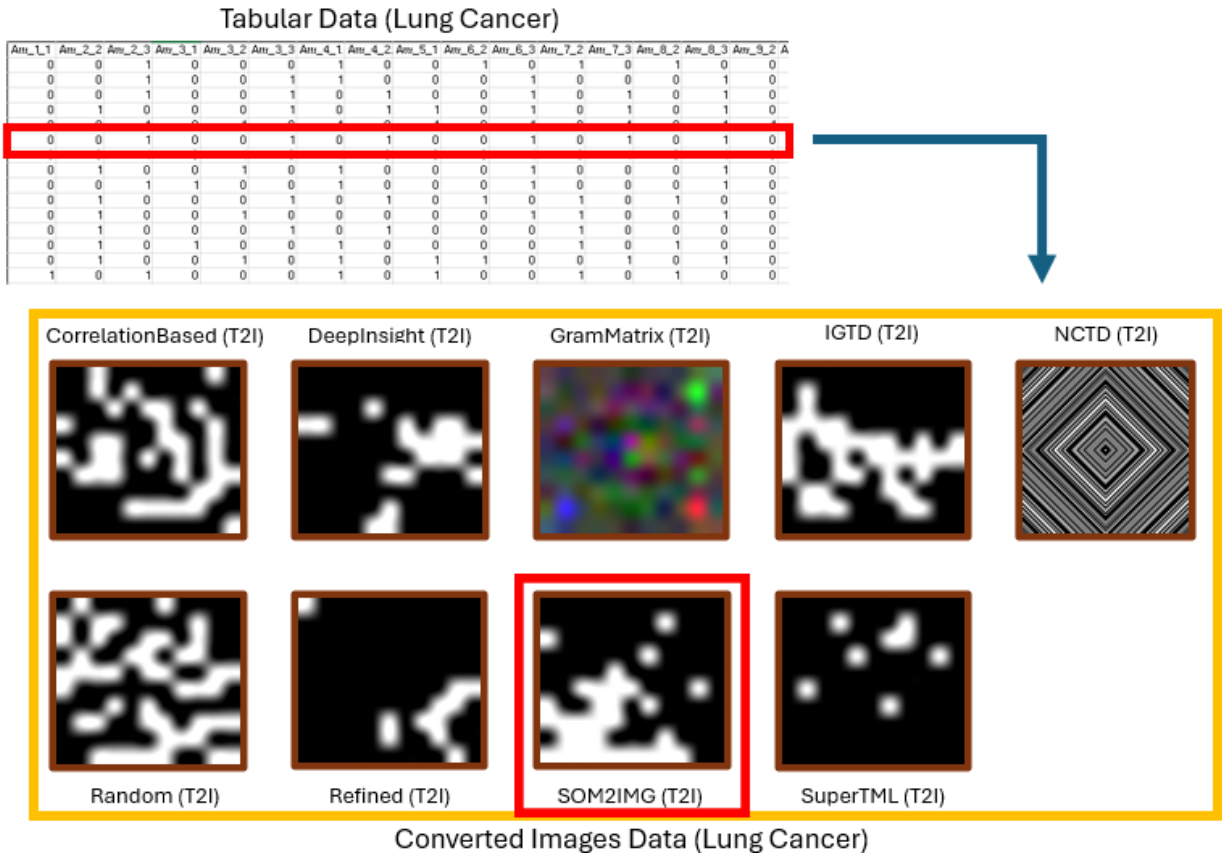


Fig. 1. Tabular-to-image (T2I) transformation methods applied to lung cancer data. Top: original tabular data matrix with binary features. Bottom: image representations from SOM2IMG, a random stacking baseline, and seven other T2I methods. Each method generates distinct visual patterns encoding tabular relationships differently.

is its (scaled) feature values at their assigned cells, preserving the original values and organizing them based on the SOM's representation.

Expanded grid (SOM2IMG-T). Collisions worsen as d increases, pushing features away from their BMUs and negatively impacting the layout. A larger grid size ($c = 3$ in Eq. 1) mitigates collisions, so we consider c as a knob and report both versions. As Sec. IV shows, the larger grid benefits continuous, higher-dimensional data but challenges small, sparse datasets, resulting in mostly empty images.

C. Compared Methods and Controls

We compare SOM2IMG and SOM2IMG-T against six existing T2I methods (DeepInsight, IGTD, REFINED, NCTD, SuperTML, and Correlation-Based mapping) and two controls: **Random** drops features at random pixels, and **SnakeRandom** places every feature once, in random order, along a space-filling path. SnakeRandom separates the value of the *arrangement* from simply keeping every feature. We drop GramMatrix, whose images barely change between samples, as it scored at chance.

D. CNN Architecture

We use one fixed CNN for every method to minimize variation due to the model architecture so that observed dif-

ferences primarily reflect the image representation. It has two convolutional blocks (16 and 32 filters of 3×3 with ReLU), global average pooling, a 128-unit dense layer with dropout, and a softmax (sigmoid for binary) output. Global average pooling lets the same network handle the very different image sizes the methods produce without blowing up the parameter count.

E. Evaluation Pipeline

To ensure fairness, all preprocessing occurs within the cross-validation loop. For each fold, we fit the min-max scaler and the T2I transform (SOM, correlation clustering, t-SNE, etc.) on the training split, build images for both splits, train the fixed CNN on the training images, and score weighted F1-score (support-weighted average of per-class F1-scores) on the held-out images. We use stratified 10-fold cross-validation, dropping to fewer folds only when a class has fewer than ten members (Lung Cancer, $N=32$, uses nine). This ensures the statistical test's mean-F1 number is not skewed by the few datasets with fewer folds.

IV. RESULTS

Table II shows the weighted F1 scores for all methods. Figure 4 compares SOM2IMG, its expanded variant, the

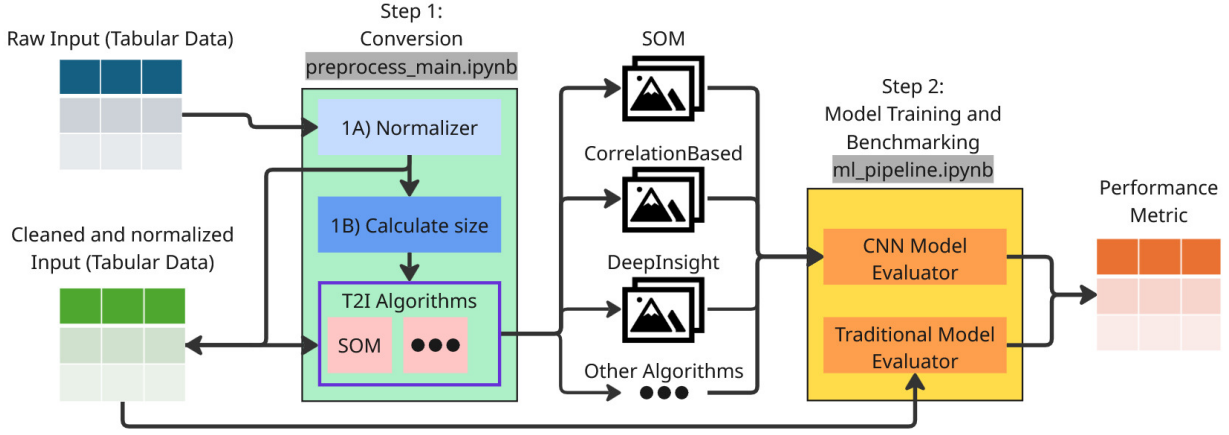


Fig. 2. Overview of the evaluation. Within each stratified cross-validation fold, the min-max scaler and every T2I transformation are fit on the *training* partition only and then applied to both partitions; the fixed CNN is trained on the resulting training images and scored on the held-out images. No information from the held-out fold informs the normalization or the feature layout.

TABLE II

WEIGHTED F1 (STRATIFIED 10-FOLD) PER DATASET; BEST PER ROW IN **BOLD**. SOM = SOM2IMG; SOM-T = EXPANDED-GRID VARIANT; SNAKER = LOSSLESS RANDOM-ORDER CONTROL. NCTD IS OMITTED ON THE TWO HIGH-DIMENSIONAL SETS, WHERE ITS $2N \times 2N$ IMAGE IS COMPUTATIONALLY INFEASIBLE.

Dataset	N	d	SOM	SOM-T	Corr	IGTD	DeepIns	NCTD	REF	STML	Rand	SnakeR
Breast Cancer	569	30	0.949	0.954	0.963	0.961	0.958	0.929	0.954	0.941	0.934	0.959
Heart Disease	303	20	0.822	0.836	0.846	0.827	0.816	0.816	0.796	0.741	0.771	0.830
Lung Cancer	32	101	0.469	0.355	0.271	0.386	0.390	0.233	0.426	0.270	0.426	0.262
Myocardial	1700	184	0.826	0.818	0.830	0.828	0.776	0.768	0.780	0.773	0.820	0.819
Heart and Soul	515	100	0.761	0.761	0.775	0.760	0.754	0.754	0.755	0.754	0.759	0.780
Obesity	2111	23	0.907	0.939	0.901	0.901	0.813	0.692	0.759	0.489	0.918	0.908
Dry Beans	13611	16	0.923	0.924	0.925	0.922	0.920	0.919	0.922	0.615	0.920	0.924
Spambase	4601	57	0.935	0.931	0.938	0.932	0.914	0.878	0.898	0.890	0.929	0.937
German Credit	1000	48	0.725	0.731	0.726	0.718	0.703	0.595	0.606	0.666	0.687	0.723
Ionosphere	351	34	0.945	0.936	0.925	0.939	0.871	0.939	0.868	0.880	0.941	0.939
MADOLON	2600	500	0.555	0.581	0.604	0.604	0.651	—	0.781	0.688	0.559	0.555
Mouse (genomics)	447	447	0.685	0.668	0.657	0.707	0.431	—	0.404	0.706	0.636	0.564

top-ranked competitor (Correlation-Based), and the random baseline. The performance differences are small. SOM2IMG achieves the highest performance on four datasets (Lung Cancer, Obesity, German Credit, and Ionosphere), while Correlation-Based is the best on five. IGTD, REFINED, and the lossless control each achieve one best result.

The impact of grid size varies by dataset: the larger grid (SOM2IMG-T) improves weighted F1 on higher-dimensional sets such as Obesity ($0.907 \rightarrow 0.939$) and German Credit, but hurts the sparse Lung Cancer set ($0.469 \rightarrow 0.355$). Grid size is thus a useful hyperparameter whose effect depends on the data, not a free improvement.

A. Statistical Analysis

We follow Demšar [12]. Since NCTD is absent on the two high-dimensional sets, we exclude it from the pooled test, resulting in $k=9$ methods present on all $N=12$ datasets. The ranks and tests use the unrounded scores. A Friedman test rejects the hypothesis that all methods perform equally ($\chi^2=28.4$, $p<10^{-3}$). The average ranks are as fol-

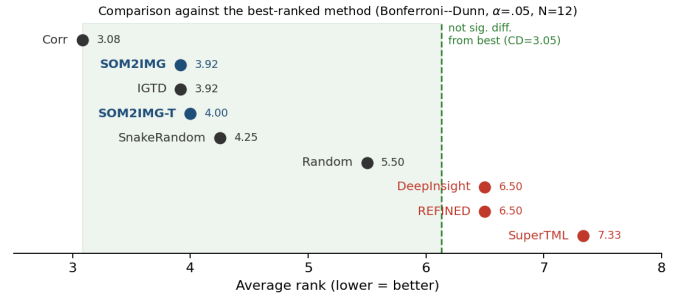


Fig. 3. Average ranks with the Bonferroni-Dunn critical difference ($CD=3.05$, $\alpha=0.05$, $N=12$, $k=9$) for comparison against the best-ranked method (Correlation-Based). Methods inside the shaded band are not significantly different from the best, so the leaders, including SOM2IMG and SOM2IMG-T (bold), form one statistically tied group with no significant winner; only the feature-dropping methods DeepInsight, REFINED, and SuperTML fall significantly short of the best.

lows: Correlation-Based (3.08), SOM2IMG and IGTD (both 3.92), SOM2IMG-T (4.00), SnakeRandom (4.25), Random

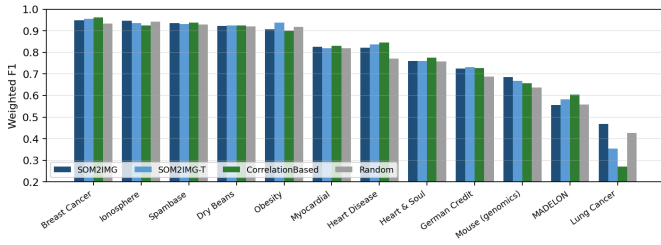


Fig. 4. Per-dataset weighted F1 for SOM2IMG, the expanded-grid variant (SOM2IMG-T), the top-ranked competing method (Correlation-Based), and the Random baseline, sorted by SOM2IMG performance.

(5.50), DeepInsight and REFINED (both 6.50), and SuperTML (7.33).

Since we are comparing everything against a single reference (the best-ranked method), we use the matching Bonferroni–Dunn/Holm post-hoc test, which sets a critical difference of $CD=3.05$ (Fig. 3). The stricter all-pairwise Nemenyi test ($CD=3.47$) only separates SuperTML. Several notable findings emerge. First, SOM2IMG and its expanded version are not significantly different from the best or from IGTD (Holm-corrected $p=1.0$); they’re in the top group. Second, DeepInsight, REFINED, and SuperTML are significantly below the best (Holm $p=0.016, 0.016, 0.001$): the same three methods that drop features (Sec. V-B). Third, the structured methods clearly rank above the random baselines, and SOM2IMG beats Random on ten of the twelve datasets, more than any other method (Correlation-Based and IGTD follow at nine). This edge does not yet reach statistical significance at the current sample size (Correlation-Based vs. Random: raw $p = 0.03$, Holm-corrected $p = 0.15$), and confirming it would likely require about twice as many datasets; the consistent direction nonetheless indicates that the learned layouts capture real structure rather than performing at chance. Lastly, this isn’t a quirk of the metric: accuracy and AUC yield the same ordering (SOM2IMG actually has the highest means, 0.807 accuracy and 0.866 AUC), and the feature-dropping methods remain at the bottom.

Across these aggregate measures SOM2IMG sits in the top tier, though per dataset it ranges from best (Lung Cancer, Ionosphere) to near the random floor (MADELON). With the leaders so close, its four per-dataset wins are best read as encouraging rather than decisive.

V. DISCUSSION

A. Performance Analysis

SOM2IMG is one of the strongest T2I methods, and the leading approaches are separated by only small margins. We read this closeness as informative rather than a weakness: strong T2I performance appears to be driven less by any particular clustering objective than by a property the leading methods share. As Sec. V-B shows, that property is preserving all features, which SOM2IMG does by construction, placing it firmly among the top performers.

B. What Distinguishes Effective T2I Methods

A clear pattern emerges from Table II and Fig. 3: methods that discard features perform worse than those that retain them. DeepInsight and REFINED lose features to coordinate-rounding collisions and SuperTML to supervised selection; all three rank significantly below the top methods, and in fact below even the random-placement baseline. The feature-preserving methods (Correlation-Based, SOM2IMG, IGTD, and the lossless SnakeRandom control) all sit in the top group. The gap widens with dimensionality: on the 447-feature genomics set the feature-droppers degrade sharply (DeepInsight 0.43, REFINED 0.40) while staying competitive on lower-dimensional data. The expanded-grid SOM2IMG variant, which cuts collisions, supports the same point.

The SnakeRandom control isolates arrangement while still preserving every feature. On low-dimensional data it matches the clustering methods, so arrangement matters little there; on the high-dimensional genomics set it falls well behind (0.564 vs. 0.657 for Correlation-Based and 0.685 for SOM2IMG), so structured placement helps more as dimensionality grows. Overall, preserving all features is the dominant factor in strong T2I performance, while the benefit of arrangement grows with dimensionality, though this last point rests on limited high-dimensional evidence and needs further validation.

C. Interpretability

SOM2IMG is also interpretable: each pixel maps to one original feature, and because the SOM places similar features together, the image itself shows how features relate. On Breast Cancer Wisconsin, for example, it groups the size features (radius, perimeter, area) and the texture features (smoothness, compactness, symmetry) into distinct regions, matching the known clinical structure without any labels.

By contrast, NCTD’s pairwise interactions and SuperTML’s glyphs obscure individual feature identities, which matters where transparency is needed.

D. Computational Considerations and Traditional Baselines

In simple terms, gradient-boosted trees remain the most effective and economical choice for tabular classification. However, T2I addresses the question of which transformation to use for a CNN pipeline. SOM2IMG emerges as a competitive and worthwhile candidate under fair evaluation. Its readable and feature-preserving layout is a significant tie-breaker since no method truly dominates. The methods also exhibit considerable cost differences. NCTD’s $2N \times 2N$ image grows quadratically with the feature count, making it impractical for our high-dimensional datasets. In contrast, the grid-based methods grow linearly in d .

E. Practical Recommendations

SOM2IMG is a strong, practical T2I option: it ranks in the top tier on every aggregate measure, beats a random baseline on more datasets than any other method, and never collapses the way the feature-dropping methods do. Its edge over the other leading methods is modest and not yet statistically

significant, so it is best seen as a dependable choice rather than a dominant one. It is especially worth reaching for when a CNN pipeline is needed and interpretability matters, or when the data is high-dimensional. Since no single transform dominates, the most reliable strategy is to try a few feature-preserving methods and keep the best, of which SOM2IMG is a competitive and uniquely readable member.

VI. CONCLUSION

We propose SOM2IMG, a novel tabular-to-image (T2I) transformation, and evaluate it across twelve datasets. The results reveal a clear top tier of feature-preserving methods, namely SOM2IMG, Correlation-Based mapping, and IGTD, that rank above the random baselines and outperform them on most datasets, while the methods that discard features fall below even a random layout. SOM2IMG is among the best of this group, achieving the top result on four datasets, the highest mean accuracy and AUC overall, and more wins over the random baseline than any other method (ten of the twelve).

These advantages are genuine but should be read with care: with only twelve datasets the comparison is underpowered, so the differences among the top methods, and their edge over a random layout, are not yet statistically significant.

A controlled ablation study further suggests a general design principle for T2I methods: preserving all features in the image representation is the dominant factor associated with strong performance, while spatial arrangement provides additional benefit primarily in higher-dimensional settings.

Future work should scale the evaluation to a larger and more diverse set of datasets. A broader benchmark, particularly with more high-dimensional genomic, text, and proteomic data, would confirm at full statistical significance that the top methods outperform random layouts and that structured placement helps in high dimensions. Additional directions include more expressive CNN architectures, adaptive grid sizing based on data characteristics, and improved handling of empty cells to reduce sparsity in low-dimensional settings.

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